ERIC CITO, BS

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EDUCATION

University of California, San Francisco (UCSF) Post-Baccalaureate **July 2025 – Present**
PROPEL: Post-Baccalaureate Research Opportunity to Promote Equity in Learning

University of California, San Diego (UCSD) **Sept 2020 – June 2025**
B.S. Computer Engineering | Minor in Cognitive Science
NCAA Division 1 Men's Tennis Student-Athlete

SKILLS

Technical: Python, PyTorch, C, C++, Java, MATLAB, R, UNIX, Bash, Git, Containers, Pandas, NeuroDSP, TensorFlow, NumPy, SimpleITK, ITK-SNAP, DICOM, CONN fMRI toolbox
Clinical: PACS, UCSF Air Radiology, APeX, MRI Safety, Human Subjects Protection
Laboratory Skills: Pipetting, Centrifugation, GC Columns, Radioactive Material Handling, Zotero, Microsoft Office Suite, LaTeX

HONORS & AWARDS

2026	International Congress of Parkinson's Disease and Movement Disorders – Travel Award
2024	IEEE Brain Discovery & Nanotechnology Workshop - Travel Award
2024	Electrical & Computer Engineering Department - Travel Award
2024	NSF California Louis Stoke Alliance for Maximizing Potential (NSF CAMP) - Travel Award
2024	NSF California Louis Stoke Alliance for Maximizing Potential (NSF CAMP) - Summer Fellowship
2024	NCAA Big West Conference Commissioner's Honor Roll
2024	UC San Diego Provost Honors – Spring Quarter
2023	UC San Diego Provost Honors – Winter Quarter
2021	NCAA UC San Diego Men's Tennis Team – Most Inspirational Award
2020	NCAA Big West Conference Commissioner's Honor Roll
2017	Weill Cornell Medicine High School Catalyst Program - First Place Poster Presentation

RESEARCH EXPERIENCE

University of California, San Francisco: *Neuromodulation Imaging Lab* **July 2025 – Present**
Title: Research Associate II (Jul 2025 – Oct 2026), Assistant Specialist III (Nov 2026 – Present)

Assistant Prof. Melanie A. Morrison (PI)
Lab Goal:
- Advancing neuromodulation therapies for patients with movement disorders and neuropsychiatric conditions. Extract quantitative information from brain MRI images to study and predict patients' neural and clinical responses to brain stimulation

Project Summaries:
- Co-developing a modular Quantitative Susceptibility Mapping (QSM) imaging pipeline to enhance visualization of deep-brain structures for support Deep Brain Stimulation (DBS) surgical planning
- Co-developing a 2D U-Net deep learning model to automatically segment implanted DBS leads to localize and visualize lead placement across brain regions
- Investigate longitudinal changes in motor network connectivity following pallidal DBS in patients with cervical dystonia using fMRI analysis
- Examine associations between whole-brain DBS lead trajectories and post-operative medication reduction and cognitive outcomes in Parkinson's disease using machine learning techniques
- Collaborating with MRI technicians to ensure high-quality acquisition of brain scans and verifying that imaging protocols are properly executed

Presented findings at:
- International Congress of Parkinson's Disease and Movement Disorders 2026
- International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting 2026
- UCSF PROPEL Symposium 2026
- UCSF Research AI Day 2026

Salk Institute of Biological Studies: *Computational Neurobiology Lab* **Nov 2023 – May 2025**
Title: Undergraduate Researcher, NSF CAMP Fellow

Prof. Terrence Sejnowski (PI)
Lab Goal:
- Advance understanding of the computational principles of the brain and develop machine-learning methods to improve diagnosis and treatment of neuropsychiatric disorders

Project Summary:
- Conducted research under Dr. Margot Wagner and Dr. Alessandra Camassa on developing quantitative diagnostic metrics for neuropsychiatric disorders using machine learning and resting-state fMRI data from the Adolescent Brain Cognitive Development (ABCD) Study (12,000 participants). Designed and implemented an outlier analysis pipeline using Deep Isolation Forest (DIF), a deep-learning–based method for robust outlier detection. Identified key sources of data variability, including regional hotspots of outliers**,** MRI vendor discrepancies, and unaccounted framewise displacement, improving model accuracy

Presented findings at:
- National Science Foundation California Alliance for Maximizing Potential (NSF-CAMP) Symposium 2025
- IEEE Brain Discovery & Neurotechnology Conference 2024
- Summer Research Conference**,** UC San Diego 2024

University of California, San Diego Cognitive Science: *Systems* Neuroscience *Lab* **Aug 2021 – Dec 2021**
Title: Undergraduate Researcher

Prof. Douglas A. Nitz (PI)
Lab Goal:
- Investigate how the brain encodes and stores spatial representations of the environment to understand the neural basis of navigation

Project Summary:
- Analyzed local field potential (LFP) recordings from hippocampal CA1 neurons in rats during maze-navigation tasks to study the dynamics of spatial memory formation
- Learned and applied signal-processing and filtering techniques to isolate burst oscillations within full LFP signals
- Contributed to research aimed at clarifying the functional dynamics of hippocampal networks and providing insight into how humans perceive and mentally map their surroundings

Memorial Sloan Kettering Cancer Center/ Weill Cornell Medicine: *Lewis Lab* **June 2018 – Sep 2018**
Title: Weill Cornell Medicine Catalyst Program Intern

Prof. Jason S. Lewis (PI)
Lab Goal:
- Develop and optimize radiotracers and antibody-based molecular imaging agents to advance cancer diagnosis and therapy

Project Summary:
- Conducted research under postdoctoral mentor Dr. Sai Sharma on HER2-targeting antibodies, studying how different levels of bioconjugation affect radiotracer synthesis and antibody immunoreactivity
- Synthesized immunoconjugates by reacting trastuzumab with increasing molar excess of isothiocyanato-desferrioxamine (p-SCN-DFO), purified products via size-exclusion chromatography, and quantified DFO-to-antibody ratios
- Found that while immunoreactivity remained stable even with 60-fold DFO excess, the association rate decreased compared to unconjugated trastuzumab, suggesting altered binding kinetics to the HER2 antigen, which could reduce therapeutic efficacy

Presented findings at:
- Weill Cornell Medicine Catalyst Program Poster Session – *First Place Award*

PUBLICATIONS & PUBLISHED ABSTRACTS

- **E Cito** et al. “Data-driven relationship between whole-brain DBS lead trajectories and medication and cognitive outcomes in Parkinson’s disease”, *ISMRM & ISMRT Annual Meeting*, Abstract Submitted, October 2027
- N Hellman, D Schoen, **E Cito** et al. “Fractal Dimension Associations with Cognitive and Mood Outcomes Following Deep Brain Stimulation in Parkinson’s Disease”, *ISMRM & ISMRT Annual Meeting*, Abstract Submitted, October 2027
- **E Cito** et al. “Longitudinal motor network responses to pallidal DBS in cervical dystonia: an ongoing DBS-fMRI study”, *Movement Disorders*, Online Supplement Published, October 2026
- **E Cito** et al. “Deep Learning-based Segmentation of Multi-target Deep Brain Stimulation Lead Trajectories on Postoperative Imaging”, *Journal of Neurosurgery*, Submitted, September 2026
- G Amouzandeh & **E Cito** et al. “Modular Framework for Automated QSM Integration into Deep Brain Stimulation Neurosurgical Planning Systems”, *Journal of Neurosurgery*, Submitted, September 2026
- **E Cito** et al. “Modular Framework for Automated QSM Integration into Deep Brain Stimulation Neurosurgical Planning Systems”, *ISMRM & ISMRT Annual Meeting*, Abstract Published, May 2026
- **E Cito** et al. “2D U-Net Segmentation of Multi-target Deep Brain Stimulation Lead Trajectories on Postoperative Imaging”, *ISMRM & ISMRT Annual Meeting*, Abstract Published, May 2026

PRESENTATIONS

- **E Cito** et al. “Longitudinal motor network responses to pallidal DBS in cervical dystonia: an ongoing DBS-fMRI study”, *International Congress of Parkinson’s Disease and Movement Disorders*, Digital Poster, October 2026
- **E Cito** et al. “2D U-Net Segmentation of Multi-target Deep Brain Stimulation Lead Trajectories on Postoperative Imaging”, *UCSF PROPEL Symposium,* Poster, June 2026
- **E Cito** et al. “2D U-Net Segmentation of Multi-target Deep Brain Stimulation Lead Trajectories on Postoperative Imaging”, *ISMRM & ISMRT Annual Meeting*, Digital Poster, May 2026
- **E Cito** et al. “Modular Framework for Automated QSM Integration into Deep Brain Stimulation Neurosurgical Planning Systems”, *ISMRM & ISMRT Annual Meeting*, Digital Poster, May 2026
- **E Cito** et al. “2D U-Net Segmentation of Multi-target Deep Brain Stimulation Lead Trajectories on Postoperative Imaging”, *UCSF Research AI Day,* Poster, Mar 2026
- **E Cito,** M Wagner, A Camassa, T Sejnowski, “Deep Isolation Forest Outlier Analysis of Large Multimodal Adolescent Neuroimaging Data” *National Science Foundation CAMP Symposium*, Poster, Feb 2025
- M Wagner, A Camassa, V Patro, K Gano, **E Cito**, G Cauwenberghs, T Sejnowski, “Deep isolation forest outlier analysis of large multimodal adolescent neuroimaging data,” Functional Imaging and Data Analysis Nano symposium, *The Society for Neuroscience*, Talk, Oct 2024
- **E Cito,** M Wagner, A Camassa, T Sejnowski, “Deep Isolation Forest Outlier Analysis of Large Multimodal Adolescent Neuroimaging Data” *IEEE Brain Discovery and Neurotechnology Workshop*, Poster, Oct 2024
- **E Cito**, M Wagner, A Camassa, T Sejnowski, “Deep Isolation Forest Outlier Analysis of Large Multimodal Adolescent Neuroimaging Data” *Summer Research Conference at UC San Diego*, Oral Presentation, Aug 2024
- **E Cito,** S Sharma, J Lewis, “The Impact of Bioconjugation on Antibody Immunoreactivity” *Weill Cornell Medicine High School Catalyst Program,* First Place Poster, Aug 2018

APPLIED ENGINEERING PROJECTS

Triton NeuroTech: *Student Software Engineer and Social Chair* **Sep 2022 – June 2023**

- Collaborated in a team to build a robotic arm using Arduino, integrating Electromyography (EMG) readings and implementing different neural networks to filter and interpret brain signals for control of robotic arm. Successfully implemented a prototype
- Coordinated and organized events to enhance the publicity of Brain-Computer Interfaces (BCI) to the public and students

Self-Driving Remote Controlled Car Racing: *AI Project* **Sep 2018 – June 2020**

- Open-source self-driving platform for small-scale remote-controlled cars
- Configured Raspberry Pi to gather operational data from steering, throttle, and camera to be used for model training, adapting from the open-source Donkey Car platform
- Deployed model on Raspberry Pi, enabling real-time autopilot functionality

- Raced in events and placed as high as second place out of 26 teams

FIRST (For Inspiration and Recognition of Science and Technology): *STEM Outreach* **Sep 2008 – June 2019**

- Part of Team Inspiration "To learn, to share, to inspire" robotics team
- Competed in Robotics competitions housing several levels from elementary school to high school including FIRST Lego League (FLL) and FIRST Tech Challenge (FTC)
- Implemented Computer Vision techniques to enable autonomous ball shooting and navigation on the robotic competition field
- Worked with 3-D printing using CAD developing robotic pieces
- Volunteered 50+ hours showcasing robotics to elementary schools, inspiring the future generation of the field of STEM
- Partnered with FIRST Global to coach team Benin as well as donated 10 robotics kits to kick start their robotics program

LEADERSHIP

UCSF PROPEL Undergrad Mentorship Program: *Mentor* **June 2026 – Present**

- Mentorship to a UCSF summer research training program student
- Provide feedback on project and final poster presentation
- Meet three times during academic year to provide ongoing mentorship on research career paths

Private Tennis Instructor: *Coach* **Sep 2020 – Present**

- Designed individualized training and strategized lesson plans for students
- Mentorship to high school students looking to participate in collegiate athletics
- Provided motivational leadership and constructive feedback to enhance performance
- San Francisco Recreation Instructor serving to teach the local community

NCAA Division 1 Men's Tennis: *Team Captain* **Sep 2021 – Mar 2023**

- Voted Team Captain for the seasons of 2021-2022 and 2022-2023
- Organized and ran captain practices as well as communication between team and coaches
- Selected for Triton Athlete Council - Student-Athlete Advisory Committee - Worked closely with athletic department to enhance the student athlete experience
- Co-started match making program to pair student-athletes with tutors struggling with STEM courses